12.3 Evaluation process (pipeline)

1. **Pre-checks (rule-based):** length bounds; language detection; content-safety classifier; PII placeholder replacement; off-topic check by comparing to scenario.
2. **Evaluation (one model call):** the model receives the scenario, response, and instructions, and returns structured JSON of observations by dimension with quoted evidence.
3. **Validation:** schema check; every quote must be found in the response; no invented details; length limits; banned phrases (e.g. "confident", "professional sounding" as a criterion).
4. **Selection:** choose up to 3 strengths and up to 2 improvements ranked by usefulness for this scenario and the user's focus.
5. **Presentation:** render into feedback cards. Raw JSON never reaches the user.
6. **Logging:** store the structured feedback, model and prompt version; never the raw audio.

12.4 Feedback dimensions

Dimension	What the AI looks for	Useful feedback looks like
Clarity	Whether the main point can be identified; terms that are unexplained for the stated audience	"Your main point, that the app helps students track assignments, comes through in your second sentence." / "You mention 'API' without saying what it does for the listener."
Structure	Whether there is an order the listener can follow (e.g. what, who for, how, where now)	"You moved from the problem to what you built. Next, you could close with where the project is now."
Specificity	Concrete details, examples, numbers, or named users rather than general claims	"You said it 'helps people'. Who specifically? One sentence about a real user would sharpen it."

Dimension	What the AI looks for	Useful feedback looks like
Relevance	Whether the response answers the prompt and scenario objective	"The question asked about a challenge. Most of your answer describes the project. Which challenge would you pick?"
Audience awareness	Whether language and depth fit the stated listener (non-technical, interviewer, peer)	"For a non-technical listener, an analogy might replace 'asynchronous requests'."
Conciseness (only where scenario asks for brevity)	Repetition or digression relative to the time guide	"You make the same point in sentences 2 and 4. Cutting one would tighten the answer."

Only the dimensions tagged in the scenario's `focus_dimensions` plus any clearly dominant issue are shown, so users do not receive six comments at once. No dimension receives a numeric grade in the MVP.

12.5 Spoken vs. typed feedback

	Spoken (transcribed, user-reviewed)	Typed
Evaluated on	The reviewed transcript only	The text as written
Not evaluated	Accent, pronunciation, dialect, pace, voice, fillers, grammar quirks of speech, punctuation produced by the transcriber	Spelling and punctuation, unless the scenario is about writing
Dimensions	Same, with listener-oriented framing ("someone hearing this")	Same, with reader-oriented framing
Extra support	Transcript is shown before evaluation with low-confidence words flagged; the user may fix words	Optional outline or bullet notes
Wording	Avoids "you wrote", says "you said"	Says "you wrote"

Transcript errors: the evaluation prompt states that transcript text may contain recognition mistakes, instructs the model never to give feedback on word choices that could be transcription errors, and never to penalise unusual words. If a user did not review the transcript, feedback includes a note that it is based on an automatic transcript.

12.6 Handling difficult inputs

Input	Behaviour
Very short (under 15 words)	"That's a start. Add a sentence about who it's for." Offer to try again. No evaluation of dimensions.
Incomplete or trailing off	Feedback on what is there and invites finishing. Never guesses the missing part.
Unclear or fragmented	Asks what the main point is, offers a restating exercise ("In one sentence: what do you want the listener to remember?").
Off-topic	Says it could not match the answer to the prompt, offers to re-read the scenario, or to practise this content under another scenario.
Mixed languages or code-switching	Evaluates meaning, not language. The MVP states that practice is evaluated in English and flags if unsupported language dominates (suggest text mode).
Copy-pasted text	Evaluated like any text. Not detected or penalised.
Inappropriate content or abuse	Respond neutrally, decline to coach, and apply rate limits.
Distress, self-harm, or safety signals	Do not coach. Respond with a supportive message and local support resources; encourage speaking to a trusted person. Log only that a safety flag occurred (no content) **[Validate with a safeguarding expert]**.
Personal data about others	Warn privately, offer to redact; do not use the content in feedback.

12.7 Rejecting feedback and trying another approach

- Each feedback item has **This doesn't fit** with reasons: *not accurate*, *not helpful*, *doesn't match what I'm aiming for*, *I'd rather keep my style*.
- **Try another way** offers a second coaching angle (e.g. "structure-first" vs. "example-first") without discarding the first.
- A **style note** setting lets a user say "Keep my informal tone" or "Keep feedback short", applied to later feedback. The AI must treat these as preferences.
- Feedback always states what is a suggestion and avoids absolute language ("You must", "Wrong").

12.8 Reducing dependence on AI-written wording

- By default, the AI never writes a full replacement answer. It gives structure patterns and short illustrative fragments for **different** content, not the user's content rewritten.
- A "Show an example approach" option is hidden by default and appears only after the user has attempted a revision. The example uses a fictional topic.
- Retry prompts ask the user a question rather than supply an answer ("What would make this concrete for a listener?").
- Progress emphasises the user's own changes, not the AI's suggestions.
- Phrases to quote back come from the user's response, not from the AI.

12.9 Guardrails (the AI must not)

Rule	Implementation
Penalise accent, dialect, multilingual patterns, non-native English	Evaluation prompt forbids; grammar and pronunciation excluded from dimensions; test set includes Nigerian-English, Indian-English, African-American English and non-native samples; failures block release
Equate fluency, extroversion or style with competence	Dimensions are task-based; no "confidence", "energy" or "presence" language; banned-term list
Require corporate polish	Prompt states multiple valid styles; tone adapts to the scenario, not a single standard

Rule	Implementation
Treat preference as failure	Subjective suggestions are labelled "one option"; objective issues (answers a different question) are separated
Invent details	Quote validator; "only refer to what the user said"; evaluation of absent elements phrased as "I didn't hear/see X"
Present scores as definitive	No overall score in MVP; no "ability" language
Guarantee outcomes	Banned phrases ("you will get the job", "interview-ready")

12.10 Presentation of feedback [Recommendation]

Use **observations, not scores.** An overall confidence or "communication score" is not recommended: it would imply a measurement the system cannot make, vary by model run, invite comparison between people, and risk discouraging users who start lower. Instead:

- Short summary line (one sentence).
- 1–3 **What worked** cards with highlighted quotes.
- 1–2 **One thing to try** cards.
- Per-dimension chips ("Structure", "Specificity") as labels for the card, not as grades.
- In Progress: a qualitative "Then and now" showing what changed, plus the user's self-rating ("How clear did this feel to you?" 1–3), which is theirs, not the AI's.

If a numeric view is later wanted [Future]: use a 3-level descriptive rubric per dimension ("Emerging / Developing / Strong for this scenario") rather than 0–100, show uncertainty, and only after the rubric is validated against human raters.

12.11 Example input and output structures

Input (to evaluation service)

```
{
  "attempt_id": "a_81f3",
  "scenario": {
    "id": "explain-project-60s", "version": 1,
    "situation": "You're at a community meetup. Someone asks what you're
working on.",
    "audience": "Friendly, curious, not technical",
    "objective": "Explain what you're building, who it helps, and where you
are.",
    "focus_dimensions": ["structure", "specificity", "audience_awareness"]
  },
  "response": {
    "mode": "voice",
    "text": "I'm building an app that helps students keep track of
assignments. It uses a database and an API so it syncs between phone and
laptop. I've finished the login and I'm working on reminders.",
    "transcript_reviewed_by_user": true,
    "low_confidence_words": []
  },
  "history": { "attempt_number": 1, "previous_focus": null },
  "user_preferences": { "feedback_depth": "brief", "style_note": "keep my
casual tone" },
  "instructions_version": "coach-v1.2"
}
```

Output (validated JSON)

```
{
  "summary": "You named the project and its stage clearly. Next, say who it
helps and why it matters.",
  "strengths": [
    { "dimension": "structure",
      "observation": "You moved from what the app does to how far along it
is, which is easy to follow.",
      "evidence": "I've finished the login and I'm working on reminders" }
  ],
  "improvements": [
    { "dimension": "audience_awareness",
      "observation": "For a non-technical listener, 'a database and an API'
may not explain what the app does for them.",
      "evidence": "It uses a database and an API",
      "suggestion": "What would a student notice? Try describing the result
instead of the tools." },
```

```
    { "dimension": "specificity",
      "observation": "'Students' is broad.",
      "evidence": "helps students keep track of assignments",
      "suggestion": "Who is the student you had in mind, and what goes wrong
for them today?" }
  ],
  "retry_prompt": "Try again with one sentence about the person you built it
for.",
  "not_evaluated": ["accent", "pronunciation", "pace", "grammar"],
  "flags": { "safety": false, "too_short": false, "off_topic": false }
}
```

12.12 Sample feedback (as the user sees it)

> **Scenario:** Explain your project in 60 seconds · *to a friendly, non-technical listener*
>
> **Your answer:** "I'm building an app that helps students keep track of assignments. It uses a database and an API so it syncs between phone and laptop. I've finished the login and I'm working on reminders."
>
> **Summary:** You named the project and its stage clearly. Next, say who it helps and why it matters.
>
> **What worked**
>
> - **Structure:** You went from what the app does to how far along it is, which is easy to follow. *"I've finished the login and I'm working on reminders."*
>
> **One thing to try**
>
> - **Audience awareness:** For someone non-technical, "a database and an API" may not say what the app does for them. What would a student notice? Describe the result instead of the tools.
> - **Specificity:** "Students" is broad. Who did you have in mind, and what goes wrong for them today?
>
> **Want to try that answer another way?** *[Revise my answer] [Try a different scenario]*
>
> *This feedback was written by AI and may be wrong. If something doesn't fit, tap **This doesn't fit**.*

12.13 Quality evaluation of the coach

- **Golden set:** 40–60 sample responses (varied accents, dialects, lengths, topics) with human-written expected observations; release gate: no fairness failures and under 10% inaccurate evidence.
- **Regression:** run the set on every prompt or model change.
- **Human review:** sample 5% of production feedback (with consented, anonymised data only) weekly in early phases.
- **Model choice:** use one capable model, with a structured output mode; avoid chaining multiple models in MVP.

13. Visual identity and design-system specification

The brand direction is a product requirement. The system below keeps the five starting colours exactly and adds only functional tints and darker shades where contrast demands it.

13.1 Brand expression

Feel: a playful digital notebook crossed with a creative studio. Whimsy lives in small details (a hand-drawn underline that draws in under a highlighted phrase, a spark when a retry improves a point), not in mascots or cartoon art. Practice and feedback screens stay calm.

13.2 Logo and wordmark

Item	Specification
Wordmark	"SpeakHER" set in Fredoka (Medium), with "Speak" in Deep Plum and "HER" in a heavier weight and Coral-to-Lilac underline stroke drawn by hand, not a different colour alone. Keep exact casing.
Detail (optional)	A small closing quotation mark replacing the dot or tail after "HER", only if it reads clearly at 24 px; otherwise omit. **[Validate in design exploration]**

Item	Specification
App icon	Rounded speech bubble in Lilac on a Warm Cream or Deep Plum tile, with a Butter Yellow spark at the upper right and a quotation-mark tail. No female silhouettes, no pink, no circuit or robot motifs.
Variants	Full wordmark (horizontal); stacked; icon only; one-colour Deep Plum; one-colour Cream (for dark backgrounds).
Clear space	Height of the "H" on all sides. Minimum width 96 px for the wordmark, 24 px for the icon.
Placement	Top-left in the header; centred on auth and onboarding; never inside practice-room content area.

13.3 Colour tokens (contrast verified)

Contrast ratios were computed against WCAG 2.2 relative luminance. Text needs 4.5:1 (3:1 for large text); UI components and focus indicators need 3:1.

Brand colour	Hex	Verified use
Deep Plum	`#35264A`	Text 13.1:1 on Cream; 6.3:1 on Lilac; 11.0:1 on Butter Yellow; 5.5:1 on Coral
Warm Cream	`#FFF9F0`	Canvas
SpeakHER Lilac	`#B9A2F6`	Surfaces and selected states only; 2.1:1 against Cream, so never as the only boundary or text
Butter Yellow	`#FFE58A`	Highlights behind Deep Plum text; 1.2:1 against Cream, so a highlight must always carry text or underline
SpeakHER Coral	`#FF7B70`	Primary button fill with **Deep Plum** text (5.5:1). **White on Coral fails (2.5:1); do not use.** Not for text or icons on Cream (2.4:1)

Semantic token	Value	Contrast and notes
`bg-canvas`	`#FFF9F0`	Page background
`bg-surface`	`#FFFFFF`	Cards on canvas (13.8:1 with plum text)
`bg-brand-soft`	`#E6DCFB`	Lilac tint for cards and selected nav; plum text 10.5:1
`bg-brand`	`#B9A2F6`	Branded surfaces; plum text 6.3:1
`bg-highlight`	`#FFE58A`	Phrase highlights and discovery; plum text 11.0:1
`bg-highlight-soft`	`#FFF3C4`	Feedback evidence quotes; plum text 12.4:1
`text-primary`	`#35264A`	13.1:1 on canvas
`text-secondary`	`#5A4A73`	7.5:1 on canvas
`text-muted`	`#6B5A85`	5.9:1 on canvas; minimum for helper text
`text-on-primary`	`#35264A`	Plum on Coral 5.5:1
`text-link`	`#5B3FC4`	6.8:1 on canvas; always underlined
`action-primary`	`#FF7B70`	Hover `#FF6A5E`, pressed `#F2574A` (plum text remains above 4.5:1; re-verify when finalised)
`action-secondary-border`	`#35264A`	2px outlined button, 13.1:1
`border-subtle`	`#E4DAF2`	Decorative dividers only (not required for comprehension)
`border-input`	`#8E7BC9`	3.4:1 on canvas; satisfies the 3:1 component rule
`focus-ring`	`#35264A` 3px, plus 2px `#FFFFFF` offset	Visible on cream, lilac and coral surfaces
`feedback-strength`	bg `#E3F4EA`, icon/border `#1F7A4D`	Border 5.1:1; always paired with an icon and label

Semantic token	Value	Contrast and notes
`feedback-improve`	bg `#FFF3C4`, border `#35264A`	Never red; improvement is not an error
`success`	`#1F7A4D`	5.1:1 on canvas
`error`	`#B8362B`	5.6:1 on canvas; white text on it 5.8:1
`warning`	bg `#FFF3C4`, text plum, with icon	
`disabled`	bg `#EFE9F5`, text `#6B5A85`	Disabled controls are exempt from contrast rules but must remain legible
Dark theme **[P1]**	canvas `#1F1630`, text `#FFF9F0`, lilac as `#B9A2F6`	Not in MVP; tokens named so a dark theme can be added

13.4 Typography

Role	Font	Size / line height (mobile → desktop)	Weight
Display (brand moments, hero, empty-state headings)	Fredoka	32/38 → 48/54	500–600
H1	Fredoka	28/34 → 36/42	500
H2	DM Sans	22/28 → 26/32	700
H3	DM Sans	18/24 → 20/28	700
Prompt (scenario text in practice room)	DM Sans	20/30 → 22/34	500
Body	DM Sans	16/26 → 17/28	400
Feedback body	DM Sans	16/26	400, max line length 70 characters

Role	Font	Size / line height (mobile → desktop)	Weight
Small / helper	DM Sans	14/20	400
Button / nav	DM Sans	16/20	600
Label / caption	DM Sans	12/16 minimum	500

Rules: Fredoka only for short expressive text (under about 8 words); never for feedback, transcripts, forms or paragraphs. Body never below 16 px. Support user zoom to 200% and system font scaling. Self-host fonts with a system-ui fallback. **[Recommendation]** Fredoka and DM Sans are retained as proposed; confirm that Fredoka's rounder forms stay credible for the older end of the audience in usability testing.

13.5 Spacing, radii, elevation

Token set	Values
Spacing (4 px base)	4, 8, 12, 16, 24, 32, 48, 64
Layout gutters	16 mobile, 24 tablet, 32 desktop; content max-width 1120; reading column max 680
Radii	8 (inputs, chips), 16 (cards), 24 (large cards, sheets), 999 (buttons, pills), speech-bubble cards use 24 with one 8 px corner as the "tail"
Elevation	`e0` none; `e1` 0 1px 2px rgba(53,38,74,.10); `e2` 0 4px 12px rgba(53,38,74,.12) for hovered cards and sheets; `e3` 0 12px 32px rgba(53,38,74,.16) for modals. Prefer 1.5 px plum outlines on cards over heavy shadows to keep the notebook feel.

13.6 Icons and graphic language

Element	Rule
Icons	24 px grid, 2 px rounded stroke, rounded caps, Deep Plum; filled variant only for the active tab. Always paired with a text label in navigation.
Speech bubbles	Containers for prompts and feedback cards; the tail points toward the speaker (the user's response bubbles tail right; the AI's tail left).
Quotation marks	Mark the user's own words in feedback evidence.
Hand-drawn marks	Underlines and circles used only to point to one phrase per feedback card; drawn as SVG paths that animate in once (200–400 ms), static under reduced motion.
Sparks and stars	Maximum one per screen, used for a meaningful moment (first completed retry, a clear improvement). Never for generic praise.
Waveform	Rounded bars, Lilac idle, Coral while recording, plum outline when paused; supplemented by text status.
Dotted paths	Subtle connectors between onboarding steps and between attempts in the history.
Illustrations	Simple line-based spot illustrations (notebook page, bubble, pencil). No human figures in MVP.
Decoration limit	None in the practice prompt area or inside feedback body text.

13.7 Components

Component	States and rules
Primary button	Coral fill, plum 600-weight text, pill, min height 48. States: default, hover (darker Coral), pressed (`#F2574A`, 1 px shift), focus (ring), disabled (grey-lilac fill), loading (spinner replaces icon, label remains). One per screen.
Secondary button	Transparent, 2 px plum outline. Tertiary: text link style with underline.
Destructive button	`#B8362B` fill, white text; always preceded by confirmation.
Icon button	44×44 target, visible tooltip and `aria-label`.
Cards	White surface, 1.5 px plum-tint outline, radius 16, e1; selectable cards show a 2 px plum border and Lilac-soft fill (never colour alone: also a check icon).
Scenario card	Title, one-line situation, audience tag, time guide, difficulty dots (with text), "Practised ×2" tag.
Form fields	Label above, helper text below, `border-input` 1.5 px, radius 8, height 48. States: default, hover, focus (ring), filled, error (icon, message, red border), disabled. Placeholder is never the label.
Chips (goals)	Toggle buttons with `aria-pressed`; selected = Lilac fill plus check icon.
Alerts and banners	Info (Lilac-soft), success, warning (yellow-soft), error. Each has an icon, a title, a message and an action. Errors use `role="alert"`.
Feedback card	Speech-bubble container; header with dimension chip; observation; evidence quote on `bg-highlight-soft` with a hand-drawn underline; action row ("This doesn't fit"). Strength cards carry a spark icon; improvement cards carry a pencil icon.

Component	States and rules
Modal / sheet	Focus trap, close on Escape, returns focus to trigger. Sheets on mobile, centred modals on desktop.
Toast	Non-critical confirmations only; 6 seconds minimum, pause on hover or focus; never the only way to learn about an error.
Progress indicator (onboarding)	Dotted path with labelled steps ("Step 2 of 5").

13.8 Audio-recording component states

State	Visual	Text status (announced)	Controls
Idle	Lilac static waveform, Coral Record button	"Ready when you are"	Record, Switch to text
Requesting permission	Button disabled, spinner	"Waiting for your browser to ask about the microphone"	Cancel, Switch to text
Permission denied	Plum-outline waveform with slash icon	"The microphone is blocked. You can type instead."	Type instead, How to enable
Recording	Coral animated waveform, elapsed time, pulsing dot (static under reduced motion)	"Recording, 00:12" (announced at start, then every 30 s, not every second)	Pause, Stop, Cancel
Paused	Waveform frozen, plum outline	"Paused at 00:12"	Resume, Stop, Cancel
Recorded	Static waveform with scrub bar	"Recorded 00:42"	Play, Re-record, Continue
Playing	Playhead moves	"Playing"	Pause, speed 0.75/1/1.25, scrub
Transcribing	Skeleton text lines	"Turning your recording into text"	Cancel

State	Visual	Text status (announced)	Controls
Transcript review	Editable text, low-confidence words underlined with a dotted line	"Check the words, then get feedback"	Edit, Re-record, Get feedback
Error	Warning banner	Reason and next step	Retry, Type instead

13.9 Other global states

State	Rule
Loading	Skeletons for lists; for AI feedback, staged text ("Reading your answer", "Finding what worked"). No fake percentage bars. Maximum 20 s before a retry option appears.
Empty	Spot illustration, one sentence, one action.
Error	Say what happened, what is saved, what to do. Never blame the user.
Success	Quiet confirmation; reserve sparks for meaningful moments.
Disabled	Explain why where practical ("Add a few more words to continue").

13.10 Responsive breakpoints

Name	Width	Layout
Mobile	360–599	Single column; bottom tab bar; sticky primary actions
Tablet	600–1023	Two-column where useful (library grid, compare view); side rail optional
Desktop	1024–1439	Left navigation rail; centred content

Name	Width	Layout
Wide	1440+	Max-width container; no stretching

Design mobile first at 360 and 390 px; test at 320 px for reflow (WCAG 1.4.10).

13.11 Accessibility standards

Target **WCAG 2.2 AA** as the design and test standard. Specifics: contrast as above; 3:1 on component boundaries and icons; visible 3 px focus ring on every interactive element; no colour-only meaning; touch targets at least 44×44 px; no time limits on responses; captions or transcripts for any audio the product plays; `prefers-reduced-motion` removes drawing animations, pulsing and parallax; text resizing to 200%; screen reader names and states for every control; support for Windows high-contrast mode via system colours on borders and focus.

13.12 How the system applies to key screens

Screen	Treatment
Landing	Cream canvas, Fredoka hero with one hand-drawn underline under "words", a speech-bubble demo card with a Butter Yellow highlight, Coral primary CTA, Lilac scenario cards. One spark maximum.
Onboarding	Centred single question per screen, dotted path progress, Lilac-soft selectable chips, plain-language privacy card with plum outline.
Dashboard	Cream canvas; one large Lilac "Suggested for you" card; white cards for recent attempts; calm, no streaks or badges.
Scenario library	Category tabs in Lilac-soft selected state; white scenario cards with plum outline; difficulty shown with dots and a text label.
Practice room	Minimal chrome. Prompt in a Lilac-soft speech bubble at the top, large editor or recorder below, Coral primary action. No sparks, no illustrations, no marketing.

Screen	Treatment
Feedback	White speech-bubble cards in short sections; Butter Yellow evidence highlights; hand-drawn underline on the single phrase in focus; a Coral "Revise my answer" button.
Progress	Dotted connecting path between attempts; side-by-side comparison with highlighted additions; no totals, ranks or streaks.

13.13 Brand voice and microcopy

Do: "Let's work on making that explanation clearer." · "You've named the project. Now explain the problem it solves." · "Want to try that answer another way?" · "You can take your time."

Don't: "Become unstoppable", "Speak like a tech queen", generic praise ("Great job!") without evidence, exclamation overload.

Privacy copy rule: use "Your practice stays private" only if it is literally true under the final data practices. A safer accurate default: "Only you can see your practice. Your answers are processed by AI and speech services to give feedback. Audio isn't kept after it's turned into text." Every privacy line must be reviewed against the implemented system before launch.